Power BI Session 1

Loading Data — First Visualization — Adding Cards — Using Slicers

Dataset: Business data
Duration: 2-3 hours — **Level:** Beginner

EXERCISE 1: LOADING AND EXPLORING DATA

Objective: Import the CSV dataset and understand the data structure.

Task:

1. Load the `business data.csv` file into Power BI
2. Examine the columns and data types
3. Answer the following questions:

Questions:

a) How many rows of data are in your dataset?

500

b) List 5 column names from your dataset and identify whether each is a dimension (category) or metric (number):

- Column: Month_name — Type: text
- Column: Region — Type: text
- Column: quantity — Type: whole number
- Column: sales_amount — Type: decimal number
- Column: Unit_price — Type: decimal number

c) What data type should the "Revenue" column have? Why?

the revenu variable should be decimal number because its a numerical var

d) What business process does this dataset represent? Describe in 2-3 sentences.

its a sales data set that contain many useful features led us to see the profitablity of our company

---

e) If you needed to analyze "which region is most profitable," which columns would you need?

i need two variables the profit and the region varilables

EXERCISE 2: CREATING FIRST VISUALIZATION

Objective: Create a column chart showing Revenue by Region.

Task:
Create a column chart with Region on the X-axis and Revenue on the Y-axis. Add an appropriate title.

Questions:

a) Which region has the highest revenue? north

b) Which region has the lowest revenue? south

c) What is the approximate revenue difference between the highest and lowest region?

49314.37

d) Why is a column chart appropriate for comparing revenue across regions? Explain your reasoning.

because its the graph that can takes both categorical and numerical vars

e) If you were the regional manager of the lowest-performing region, what questions would you ask based on this chart?

how we can increase the revenu in south region ?
what is the best methodes for achieveing our goals

f) If you swapped the axes (putting Revenue on X-axis and Region on Y-axis), would the chart still be useful? Why or why not?

yes for sure it still be useful , because it will be a bar shart the same as column shart

EXERCISE 3: CREATING KPI CARDS

Objective: Create three summary cards displaying key metrics.

Task:
Create three separate card visualizations showing:

1. Total Revenue (across all orders)
2. Total Profit (across all orders)
3. Total Number of Orders

Format each card with a clear label and appropriate currency/number formatting.

Questions:

a) What is the Total Revenue for all orders?

1.12M

b) What is the Total Profit for all orders?

606.3K

c) How many total orders are in the dataset?

500

d) Calculate the Average Order Value (Total Revenue ÷ Total Orders):

2.23k

e) Calculate the Profit Margin (Total Profit ÷ Total Revenue × 100):

54.29%

What does this percentage tell you about business profitability?

it tels us that we reach 54.29 $ profitability

f) If the company's goal is a 25% profit margin, are they meeting this goal? Explain.

yes we met the goal 54.29% >25%

g) Rank these three KPIs in order of importance for business decision-making (1 = most important, 3 = least important). Justify your ranking:

- Revenue: __2____
- Profit: __1____
- Orders: ___3___

EXERCISE 4:USING SLICERS FOR INTERACTIVITY

Objective: Add interactive filters (slicers) to enable dynamic analysis.

Task:
Create three slicers for:

1. Region (dropdown style)
2. Product_Category (dropdown style)
3. Sales_Channel (buttons or dropdown)

Connect the slicers to your visualizations from Exercises 2 and 3.

Questions:

a) Use the Region slicer to filter for only the "North" region:

- What is the total revenue for North region only?

  244.20K

- How many orders are in the North region?

  109

b) Now select only "Electronics" in the Product_Category slicer:

- How did the revenue change when you added this filter?

  238.34K

- Which region-product combination is most profitable?

  west

c) Filter for "Online" sales channel:

- Is Online a more or less profitable channel compared to Retail?

  online profit 144.2K retail profit 149.5K online sales_channel is less profitable than retail

- How does the order count differ between Online and Retail?

  online order 132 retail order 126

d) Select multiple regions (e.g., North AND South):

- What is the combined revenue for these two regions?

  439.08K

- Can slicers filter for multiple values at once? Explain your answer.

  yes we can select multiple choices , by modifying the settings of slicers

e) Clear all filters and explain:

- What is the purpose of having slicers on a dashboard?

  to make the dashboard easy to read

- How do slicers improve the user experience compared to looking at a static table?
  Slicers make the dashboard interactive. Users can quickly filter and explore data (by date, product, region, etc.) instead of manually searching through a large static table. This makes analysis faster, clearer, and easier to understand.

- Who would benefit from using slicers on this sales dashboard?

  manager , sales teams ,analysts

EXERCISE 5: COMPREHENSIVE ANALYSIS

Objective: Use all tools from previous exercises to analyze business performance.

Task:
Using your visualizations, slicers, and KPI cards, answer the following business questions:

Questions:

a) **Regional Performance:**

Without any filters, rank the 3 regions by revenue (highest to lowest):

1. north
2. west
3. east

b) **Product Category Analysis:**

Use the Product_Category slicer and check each category individually

- Which product category generates the most profit?

  beauty profit 130.5K ,clothing profit 123.3K, electronics profit 129.2k , home & garden profit 117.5K, sports profit 105.7K we see that beauty product cat generates the most profit

- Which product category has the lowest profit?

  sports product category has the lowest profit

c) **Sales Channel Comparison:**

Compare Total Revenue across all four sales channels (Online, Retail, Wholesale, Direct)

- Which channel generates the most orders?

  direct 109 , online 132 , retail 126 , wholesale 133 wholesalee generates the most orders

- Which channel generates the most profit?

  direct p 133K , 144.2 k , retail 149.5K, wholesale 179.7k wholesale generates the most profit

d) **Regional + Product Analysis:**

Set Region = "East" and Product_Category = "Electronics"

- What is the revenue for this specific combination?

  38.85K

- How many orders in this combination?

16 order

e) **Filtering to Find Insights:**

Apply the filter: Product_Category = "Clothing" AND Sales_Channel = "Online"

Compare this to: Product_Category = "Clothing" AND Sales_Channel = "Retail"

Which channel is more effective for clothing sales?

online clothing sales have profit of 31.3K

retail clothing sales have a profit of 27.2K so the best is online

f) **Business Recommendation:**

Based on all your analysis, if you could invest in improving ONE aspect of the business, what would it be?

i will invest more in the south region and improve more my business for retail sales channel

Support your answer with 2-3 data points from your visualizations:

CHALLENGE QUESTIONS (OPTIONAL)

If you finish early, try these additional questions:

1. Create a new visualization showing Revenue by Product_Category. Which category is most valuable?

   _____

2. Create a second visualization showing the count of Orders by Sales_Channel. How are orders distributed across channels?

   _____

3. Using both visualizations together, identify the product-channel combination with the highest revenue and explain why this might be important.

   _____

4. If you had to choose to eliminate one product category due to budget constraints, which one would it be? Use your data to support your decision.

   _____

5. Compare the profit margin across different regions. Are some regions more profitable than others even if they have similar revenue?

   _____

SUBMISSION CHECKLIST

Before submitting your work, verify you have completed:

Exercise 1: Data loaded and explored with all questions answered

Exercise 2: Column chart created with questions answered

Exercise 3: Three KPI cards created with questions answered

Exercise 4: Three slicers created and connected with questions answered

Exercise 5: Comprehensive analysis with business recommendations

Challenge Questions: Attempted (optional)

Report is clean, organized, and easy to understand

Good Luck!
Happy Learning!